

Standard Operating Procedure

RS-485 Communication Bus Design & Installation — Kantech Access Control

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Related Documents	SOP-LV-010, TSW-LV-001, REF-AC-RS485-001
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Applies To	Field Technicians, Installers
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Owner	Apex – Operations
Approved By	_____

NOTE ON THIS DOCUMENT:

The RS-485 electrical specifications below are drawn from the TIA/EIA-485-A standard (the governing standard for RS-485, maintained by the Telecommunications Industry Association). Kantech-specific figures (max loop length, recommended cable) are drawn from Kantech's published KT-400 installation documentation. This SOP covers install-time bus design — for diagnosing a bus that is already installed and misbehaving, see TSW-LV-001.

1. Purpose

This procedure establishes the standard for designing and installing RS-485 communication loops connecting Kantech KT-series controllers, to prevent the communication faults, dropouts, and troubleshooting call-backs that result from incorrect topology, termination, or cable selection at install time.

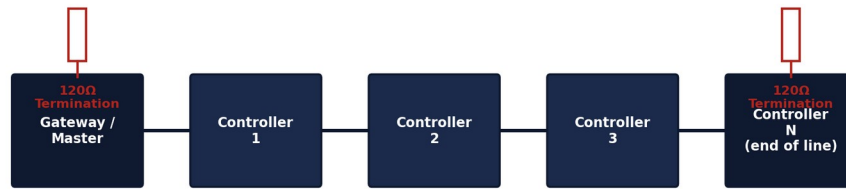
2. Scope

Applies to all Apex technicians designing, installing, or extending an RS-485 controller bus on a Kantech EntraPass deployment.

3. Topology — Daisy Chain Only

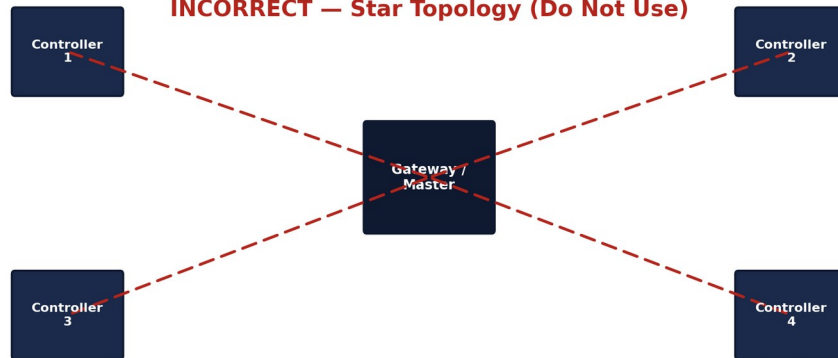
- RS-485 must be wired as a linear daisy chain (bus) with exactly two physical ends — controller to controller to controller, in sequence.
- Do not wire in a star topology (multiple branches from a single point) or a ring — both cause signal reflections and unreliable communication, even when the bus otherwise appears to work under light load.
- Keep any stub (a short branch off the main trunk to reach a controller) as short as physically possible. Long stubs are a common source of intermittent, hard-to-diagnose communication faults.
- Where a site's physical layout makes a true daisy chain difficult (e.g., multiple electrical rooms in different directions from a central point), document the actual routing on the site wiring reference (see REF-AC-RS485-001 for the format) so future troubleshooting isn't guessing at topology.

CORRECT — Linear Daisy Chain



Termination enabled ONLY at the two physical end devices. Mid-chain controllers: termination OFF.

INCORRECT — Star Topology (Do Not Use)



Multiple branches from a single point cause signal reflections and unreliable communication, even under light load. Reference: SOP-LV-013, Section 3.

4. Cable Selection

- Use a twisted-pair cable rated for RS-485 — Kantech's own documentation specifies Ethernet Category 3 (or better) twisted pair for the communication loop.
- A dedicated signal ground/reference conductor is required in addition to the twisted pair — RS-485 is a 3-wire system (A/+, B/–, and a ground reference), not just a 2-wire pair.
- Use shielded cable in areas with elevated electrical noise (near large motors, transformers, or fluorescent/high-voltage lighting runs); ground the shield at one end only — see Section 6.

5. Loop Length & Device Count

- Maximum RS-485 loop length for a Kantech controller bus is 1.2 km (4,000 ft) using appropriate cabling, per Kantech's published specification.
- The TIA/EIA-485-A standard limits a single segment to 32 unit loads. Most Kantech controllers present as a standard (1) unit load — confirm against the specific controller model if a loop is approaching this limit.
- Where a site's controller count or cable run would exceed these limits, split into multiple loops off separate gateway/NCC ports rather than pushing a single loop past its rated length or device count.

6. Termination & Bias

- Install a termination resistor (120Ω, matching the cable's characteristic impedance) at each of the two physical ends of the bus — and only at the two ends.
- Do not enable termination on any mid-chain controller — check the controller's termination jumper/DIP switch position on every unit as part of installation, not just the ones you expect to be end-of-line.
- Do not duplicate termination or bias resistors across multiple controllers on the same loop — this is a common cause of signal degradation that can be intermittent rather than a hard failure.

7. Grounding

- Ground the RS-485 shield (if used) at one point only — typically at the gateway/panel end. Grounding at multiple points creates a ground loop and can introduce noise onto the data pair.
- For a loop spanning multiple buildings or electrical services, check for ground potential difference between the buildings before connecting the run — an isolator may be required.
- Confirm controller chassis grounds are bonded per code and not left floating.

8. Installation Practices

- Keep the untwisted portion of wire at each termination point as short as possible — excessive untwisted length reintroduces the noise susceptibility the twisted pair is meant to prevent.
- Strip insulation to the minimum length needed at each terminal; avoid over-tightening terminal screws, which can nick or break conductor strands.
- Label both ends of every run at the time of installation — not after the fact. Include the run on the site wiring reference document (REF-AC-RS485-001 format) as it's installed, not from memory afterward.
- Avoid running the RS-485 cable in the same conduit or bundle as line-voltage wiring or noisy loads (ballasts, motors, VFDs).

9. Commissioning Checks

- Before closing up any enclosure, verify termination is enabled on only the two true end-of-line controllers for that loop.
- Confirm every controller on the loop shows Online status in EntraPass before considering the installation complete.
- Where available, verify end-to-end loop resistance and voltage at the last panel on the loop — record these baseline readings on the site file. Comparing against this baseline is the fastest way to spot degradation later (see TSW-LV-001, Section 4, Per-Loop Electrical Log).
- Walk the physical run and confirm it matches the as-built documentation — do not rely on the design drawing alone if field conditions required deviation.

10. References

- TIA/EIA-485-A — Electrical Characteristics of Generators and Receivers for Use in Balanced Digital Multipoint Systems
- Kantech KT-400 Ethernet Four-Door Controller Installation Manual
- SOP-LV-010 — KT-1/KT-2/KT-4/KT-400 Controller Default & Configuration
- TSW-LV-001 — RS-485 Communication Loop Troubleshooting Worksheet
- REF-AC-RS485-001 — site wiring reference format